PHI1
PHI2-INPUT
1 MHz Clock input from 1541 System board
R/W
RES
VCC
A2
A11-EXT
FETCH-2MHZ
Set Clock Speed
RAM R/W
EN-2MHZ
Select DOS
A1
/EN-A(4..7)-AH
AH[4..7]
AL[0..3]
A[0..15]
D[0..7]
ROM-CE
ROM-BANKSEL
RAM-CS1
PHI2-IN
PHI2-OUT
GND
GND2

Alternate values for R1/C1 & R2/C2 are
R1 & R2 = 5K6
C1 & C2 = 100 pF

The main changes between Professional DOS 1.02 and Disk Demon are as follows:

1) EPROM changed from 27128 to 27256.
The main reason for the change in size appears to be because the Disk Demon ROM contains a complete copy of the original Commodore 1541 DOS.
This ROM is selected by the @D0 command from the Commodore 64

2)The circuitry for the 2Mhz clock generator has been simplified. A14 & A1 are no longer used to enable it
U6A is now used to switch between CBM 1541 DOS and Disk Demon DOS.
U6B is used to enable 2Mhz

3) The values of R1, R2, R3 & R4 are different, slightly shortening the pulse duration generated by U2

Rev 2 Additional modifications
The original R1 & R2 have been replaced by a single 10k resistor (R1)
The original R3 & R4 have been replaced by a single 10k resistor (R2)
U1, 82S123 PROM has been replaced with an ATF16V8 PLD

Rev 3 – Additional modifications
U3 moved & pins renumbered
U1 moved from under the CPU to simplify access and assembly
Diodes D1 & D2 moved from edge of PCB
Added bypass Capacitors C4, C5 & C6, 100nF
Unused inputs for U3C & U3B no longer open circuit

Rev 4
Corrected error in Schematic
U6, Pin 1 (CLR) & Pin 13 (CLR) now connect to RES
U6, Pin 4 (PRE) & Pin 10 (PRE) now connect to R7 (1K)
Added bypass Capacitor C7, 100nF
Added Ground point for scope

6502 header pins 1 & 21 (Vss) are joined by 1541 system board

PHI2-IN
1-2 MHz Clock to 6502

Unofficial Disk Demon 2026 Rework
Updated Disk Demon based on examination of an original Datel Disk Demon PCB
Original Professional DOS 1.02 by Wolfgang Moser

Niall Edwards

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Title: Disk Demon 2026 Rework

Size: A3	Date: 2026-09-08	**Rev: 4**
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